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Chapter 2: Polynomials

Every formula and result of the chapter on one sheet

1	Linear polynomial	$ax + b$, $a \neq 0$; zero $= -\frac{b}{a}$
2	Quadratic polynomial	$ax^2 + bx + c$, $a \neq 0$; at most 2 zeroes
3	Cubic polynomial	$ax^3 + bx^2 + cx + d$, $a \neq 0$; at most 3 zeroes
4	Sum of zeroes (quadratic)	$\alpha + \beta = -\frac{b}{a}$
5	Product of zeroes (quadratic)	$\alpha\beta = \frac{c}{a}$
6	Quadratic from its zeroes	$k\,[x^2 - (\alpha + \beta)x + \alpha\beta]$, $k \neq 0$
7	Sum of squares of zeroes	$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
8	Sum of reciprocals	$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$
9	Square of the difference	$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$
10	Sum of cubes of zeroes	$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
11	Cubic: sum of zeroes	$\alpha + \beta + \gamma = -\frac{b}{a}$
12	Cubic: sum of products in pairs	$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$
13	Cubic: product of zeroes	$\alpha\beta\gamma = -\frac{d}{a}$
14	Graph of a quadratic	Parabola; opens upwards if $a > 0$, downwards if $a < 0$